Equalizer & Amplifier

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Abstract

This report documents the comprehensive design, simulation, and implementation of a complete analog audio processing circuit, featuring a three-band active equalizer and an integrated power amplifier. The primary objective was to develop a functional system capable of accepting a standard line-level audio signal, allowing for detailed tonal shaping of bass, midrange, and treble frequencies, and delivering an amplified output suitable for driving a small speaker.

The system architecture follows a modular, multi-stage signal path. An initial preamplifier stage, built around a TL074 operational amplifier in a non-inverting configuration, provides a high-impedance buffer for the input source and applies a small, fixed gain to condition the signal. The core of the circuit is a parallel filter network where the signal is split into three bands. Each filter utilizes a second-order Sallen-Key topology, chosen for its stability and effective -12 dB/octave roll-off slope. Crossover frequencies were established at approximately 250 Hz and 4.1 kHz, with the band-pass (midrange) filter being realized by cascading a high-pass and a low-pass filter. Independent level control for each band is achieved using 10 kΩ audio-taper potentiometers, allowing for a customized mix. These processed signals are then recombined by an inverting summing amplifier configured for unity gain.

The final output stage consists of a master volume control feeding into an LM386, a dedicated low-power amplifier IC, which provides the necessary current gain to drive a speaker. The design accommodates a split ±9V supply for the signal processing op-amps to ensure adequate headroom and a single +9V supply for the power amp, with a DC-blocking coupling capacitor at the final output. The entire circuit's performance was rigorously verified in LTspice using both transient and AC analysis. Simulations confirmed the filter responses, signal summation, and the final amplification stage's behavior, using both sine waves and complex audio signals from imported .wav files. The project concludes with the generation of industry-standard Gerber and drill files from a completed PCB layout in KiCad, rendering the design ready for physical fabrication.

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1 Project Introduction

Audio equalization is a fundamental technique in sound engineering, used to adjust the balance between different frequency components within an audio signal. From high-fidelity home stereos to professional recording studios, the ability to shape the tonal character of sound by boosting or cutting specific frequency bands is essential. This project documents the complete design and simulation of a three-band active audio equalizer and amplifier, built from the ground up.

The objective is to create a fully functional analog circuit that takes a standard line-level input, separates it into bass, midrange, and treble frequency bands, allows for independent level control of each band, and recombines the signals to drive a speaker. The design follows a modular, multi-stage approach, beginning with a high-impedance preamplifier to buffer the input signal. The core of the circuit is a parallel filter network employing second-order Sallen-Key active filters to ensure a clean and effective separation of frequencies. The project utilizes common and well-regarded integrated circuits, including the TL074 quad op-amp for signal processing and the LM386 power amplifier for the final output stage.

This report will detail each stage of the circuit, from the theoretical design and component value calculations to the practical verification of its performance using LTspice simulation software. The process demonstrates a complete workflow, concluding with the final steps required for printed circuit board (PCB) fabrication, providing a comprehensive guide for creating a functional and customizable piece of audio hardware.

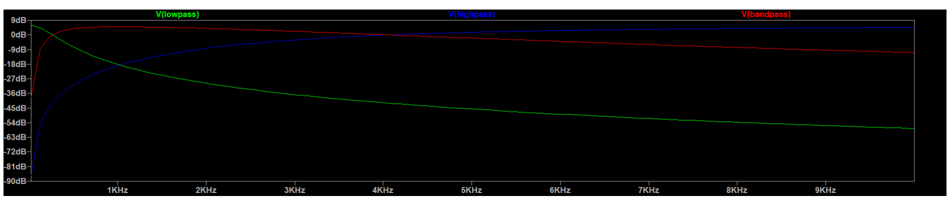

Figure 1: Simulated frequency response of the three active filter stages, illustrating the separation of the audio spectrum into distinct low-pass (bass), band-pass (midrange), and high-pass (treble) bands.

2 Circuit Design & Theory

2.1 System Architecture & Flow

The signal path begins at the 3.5mm TRS input jack, where a single channel is selected to provide a mono audio source. This line-level signal is immediately routed to a high-impedance preamplifier stage. Implemented with a TL074 op-amp, this stage serves as a buffer to prevent loading the audio source and applies a small, fixed gain to ensure a robust signal level for processing.

Following the preamplifier, the signal is split and fed in parallel to a three-band active filter network. This network consists of three independent second-order Sallen-Key filters: a low-pass filter to isolate the bass frequencies, a band-pass filter for the midrange, and a high-pass filter for the treble. The output from each of these filters is then directed to a dedicated potentiometer, which provides independent level control for its respective band.

The three adjusted signals are subsequently recombined into a single mono signal by an inverting summing amplifier. Finally, this fully mixed signal passes through a master volume potentiometer before entering the LM386 power amplifier. This final stage provides the necessary current gain to drive a speaker, delivering the audible output.

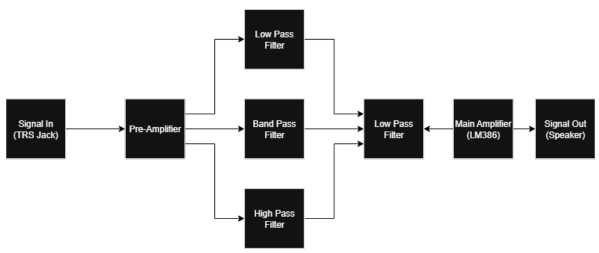

Figure 2: High-level system architecture, showing the signal path from the input preamplifier, through the parallel filter bank, to the summing and power amplifier stages.

2.2 Pre-Amplifier Stage

The first stage of the signal path is a unity-gain preamplifier, designed solely to buffer the incoming audio signal. This stage is crucial for providing a stable, high-impedance load to the audio source (e.g., a phone or computer), preventing signal degradation or a change in tonal character.

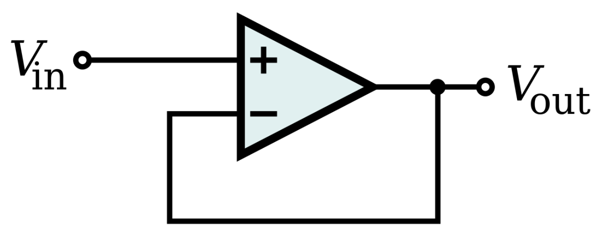

Figure 3: Voltage Follower / Buffer [2]

The circuit is implemented as a voltage follower using a single op-amp from a TL074 integrated circuit. In this configuration, the op-amp's output is connected directly to its inverting input, resulting in a precise voltage gain of 1. This ensures the signal is passed to the filter network without any change in amplitude. An additional 100 kΩ pull-down resistor is included at the input to ensure the amplifier's input is tied to a ground reference if the audio source is disconnected.

2.3 Active Filter Network

2.3.1 Filter Topology

For the active filter network, the Sallen-Key topology was selected due to its excellent balance of performance, stability, and simplicity.

This second-order filter design provides a sharp -12 dB per octave roll-off slope while requiring a minimal number of components: a single operational amplifier, two resistors, and two capacitors per filter.

The Sallen-Key is a non-inverting topology known for its high stability. Because the signal is fed into the op-amp's high-impedance non-inverting input, the filter does not load the preceding preamplifier stage. Furthermore, the op-amp is configured as a unity-gain buffer, which provides a low-impedance output to drive the subsequent control stage without adding unwanted gain. These characteristics make the Sallen-Key an ideal and practical choice for this audio equalizer project, offering reliable performance without excessive circuit complexity.

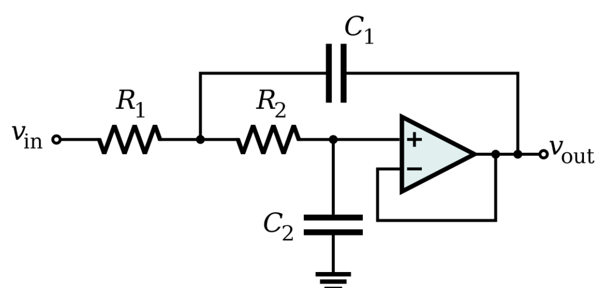

Figure 4: The Sallen-Key active filter topology used for the equalizer. This unity-gain, non-inverting design provides a stable second-order (-12 dB/octave) response with a minimal component count.[4]

2.3.2 Low-Pass Filter Design

The low-pass filter is designed to isolate the bass frequencies of the audio signal. A cutoff frequency (f_c) of approximately **250 Hz** was chosen to define the upper limit of this band. Frequencies below this point are passed through to the output, while frequencies above are attenuated.

The filter is implemented using a unity-gain Sallen-Key topology. Based on the target cutoff frequency, the following component values were selected:

- Resistors (R_1, R_2): 6.2 kΩ
- Capacitors (C_1, C_2): 0.1 μF (100nF)

These values yield a calculated cutoff frequency of approximately 257 Hz, which is ideal for the bass section of the equalizer.

2.3.3 Band-Pass Filter Design

The band-pass filter is designed to isolate the midrange frequencies, which are critical for the character of most instruments and vocals. This filter was realized by cascading a high-pass filter stage with a low-pass filter stage. This method defines a passband between the cutoff frequencies of the two filters.

To set the lower frequency boundary, a **high-pass filter** with a cutoff frequency (f_c) of approximately **250 Hz** was designed. Its component values are:

- Resistors (R_1, R_2): 6.2 kΩ
- Capacitors (C_1, C_2): 0.1 μF (100nF)

To set the upper frequency boundary, a **low-pass filter** with a cutoff frequency (f_c) of approximately **4.1 kHz** was designed. Its component values are:

- Resistors (R_1, R_2): 3.9 kΩ
- Capacitors (C_1, C_2): 0.01 μF (10nF)

By passing the signal through these two filters in series, only the frequencies between approximately 250 Hz and 4.1 kHz are passed to the output. This implementation requires two op-amp stages.

2.3.4 High-Pass Filter Design

The high-pass filter is designed to isolate the treble frequencies, which provide clarity and "air" to the audio signal. A cutoff frequency (f_c) of approximately **4.1 kHz** was chosen to define the lower limit of this band. Frequencies above this point are passed through to the output, while frequencies below are attenuated.

The filter is implemented using a unity-gain Sallen-Key topology. To achieve the desired cutoff frequency, the following component values were selected:

- Resistors (R_1, R_2): 3.9 kΩ
- Capacitors (C_1, C_2): 0.01 μF (10nF)

This combination provides a sharp and effective separation between the midrange and treble frequencies.

2.4 Summing & Control Stage

After the audio signal is separated into three distinct frequency bands, the Summing and Control Stage is responsible for allowing user adjustment of each band's level before mixing them back into a single mono signal. A schematic of this stage is shown in Figure 5.

Control is achieved using three **10 kΩ audio-taper potentiometers**, one for each filter output. Each potentiometer is wired as a **variable voltage divider**, which acts as an independent volume control for the bass, midrange, or treble band. This allows the user to attenuate each band anywhere from 100% down to 0% of its level.

The three adjusted signals are then fed into an **inverting summing amplifier**, implemented with a single TL074 op-amp. Each input signal is connected to the inverting terminal through its own 10 kΩ input resistor. A 10 kΩ feedback resistor connects the output to the inverting terminal, setting the gain for each channel to -1 ($G = -R_f/R_{in}$). This unity-gain configuration ensures that the bands are mixed equally and predictably based on the potentiometer settings. The resulting output is a single signal that represents the sum of the three frequency bands, phase-inverted by the amplifier, which is inconsequential for the final audio output.

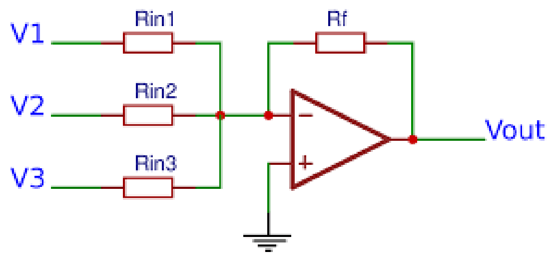

Figure 5: A three-input inverting summing amplifier.[1]

2.5 Power Amplification Stage

The final stage of the circuit is a power amplifier, designed to take the line-level output from the equalizer and provide the necessary current gain to drive a speaker. For this task, the **LM386** integrated circuit was chosen due to its simplicity and suitability for low-power audio applications. A schematic of the basic implementation is shown in Figure 6.

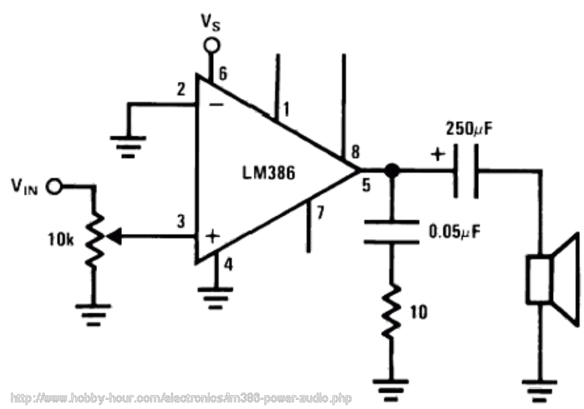

Figure 6: Basic LM386 Power Amplifier Circuit Schematic [3]

The mixed signal from the master volume control is fed into the non-inverting input (Pin 3) of the IC. The amplifier's gain is internally set to a default of 20, which provides adequate amplification for a line-level signal.

A critical aspect of this stage is the handling of the output signal. Because the LM386 operates on a single power supply (e.g., +9V and Ground), its output pin is internally biased to approximately half the supply voltage. To prevent this DC offset from reaching the speaker, which could cause damage and prevent sound production, a large **220 μF** electrolytic capacitor is placed in **series** with the output. This **coupling capacitor** blocks the DC component while allowing the amplified AC audio signal to pass through to the speaker, completing the signal path.

3 Simulation & Verification

3.1 Software & Models

All circuit design and performance verification were conducted using LTspice, a high-performance SPICE simulation software. While standard passive components were sourced from the default libraries, the specific integrated circuits central to this design required the use of external models to ensure an accurate simulation.

For the signal processing stages, the TL074 quad operational amplifier was used. A PSpice model for this component was sourced directly from the manufacturer, Texas Instruments. For the final power amplifier stage, a well-established community-provided SPICE model for the LM386 integrated circuit was imported.

The use of this software and these specific models was critical for verifying the calculated frequency response of the filters and confirming the overall functionality of the complete circuit prior to physical prototyping.

3.2 Frequency Response Analysis

To verify that the filters perform according to their design specifications, an AC analysis was conducted in LTspice. This simulation generates a Bode plot, which shows the circuit's gain in decibels (dB) across the audio frequency spectrum, proving the functionality of the equalizer.

3.2.1 Individual Filter Responses

First, each filter section was simulated independently to observe its unique frequency response. The resulting plot, shown in Figure 7, overlays the outputs of the low-pass, band-pass, and high-pass filters. The low-pass filter shows a flat passband before rolling off at its cutoff frequency of approximately 257 Hz. Conversely, the high-pass filter attenuates low frequencies and establishes its passband above the 4.1 kHz cutoff. The band-pass filter clearly demonstrates a passband between these two crossover points. These results confirm that each filter successfully isolates its intended frequency range.

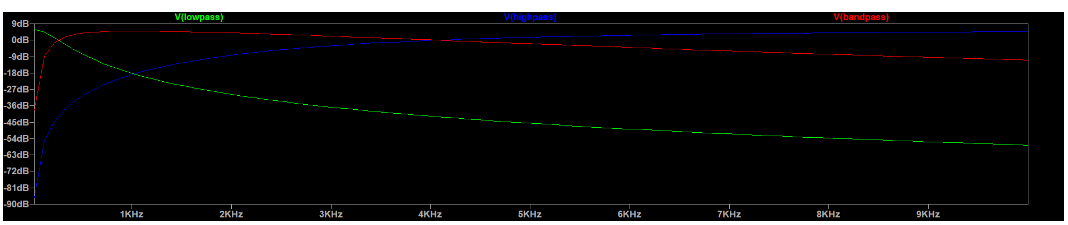

Figure 7: A Bode plot showing the individual frequency responses of the low-pass, band-pass, and high-pass filters.

3.2.2 Total System Response

Next, an AC analysis was performed on the final output of the summing amplifier, with all three level controls set to maximum. The resulting plot, shown in Figure 8, demonstrates the frequency response of the entire system. The plot is a relatively flat line across the audible spectrum (20 Hz to 20 kHz), free of significant peaks or dips. This is the desired outcome, as it proves that the individual filter bands blend together smoothly and that the equalizer does not unintentionally color the sound when its controls are in a neutral position.

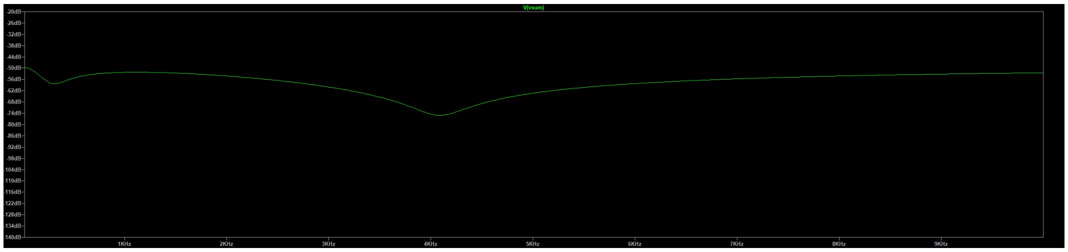

Figure 8: The frequency response of the final summed output, showing a relatively flat response across the audible spectrum.

3.3 Transient Analysis

To complement the frequency response data, a **transient analysis** was performed to verify the circuit's real-world performance with a complex audio signal. This simulation models the circuit's behavior over time, showing the actual output waveform as if it were processing a real song.

For this test, a short (3-second) `.wav` audio file was used as the input to the preamplifier. The simulation was run for the full duration of the clip, and the output was captured at two key points: after the summing amplifier (line-level output) and after the LM386 power amplifier (speaker-level output).

The resulting plots, shown in Figure 9, display the input and output waveforms over a short time slice. The output signal correctly follows the contours of the input, demonstrating that the circuit amplifies and processes the complex waveform without introducing obvious distortion or instability. This successful transient analysis confirms that the equalizer functions correctly not just with simple sine waves, but with dynamic, real-world audio signals.

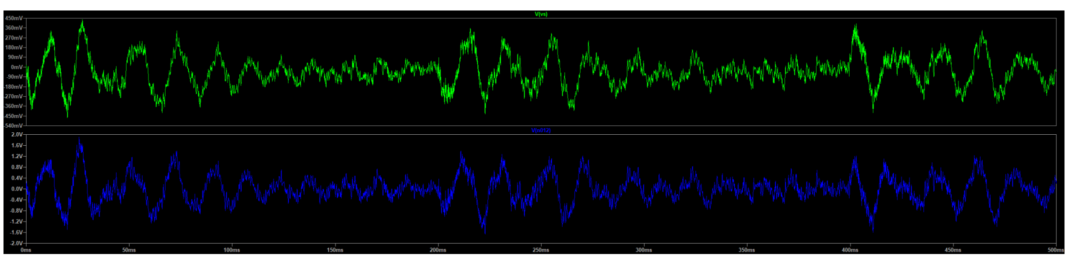

Figure 9: A transient analysis plot showing the input audio waveform from the `.wav` file (top) and the final amplified output waveform (bottom) over a 20ms time slice.

4 PCB Layout & Fabrication

4.1 PCB Design

The final stage of the project was to translate the verified schematic into a physical printed circuit board (PCB) layout using KiCad design software. A two-layer board was chosen for this design as it provides an excellent balance of performance, simplicity, and cost-effectiveness for a mixed-signal audio project of this complexity.

The component placement was organized logically to mirror the signal flow, with the input stage on one side of the board and the power amplifier on the other to minimize noise coupling. Signal traces were kept as short and direct as possible. To ensure a low-impedance return path for all signals and to shield the analog circuitry from interference, a ground plane was implemented on both the top (`F.Cu`) and bottom (`B.Cu`) layers using KiCad's "Add a filled zone" tool. These two planes were connected, or "stitched," together at numerous points using vias to create a unified ground network, as shown in the layout in Figure 10.

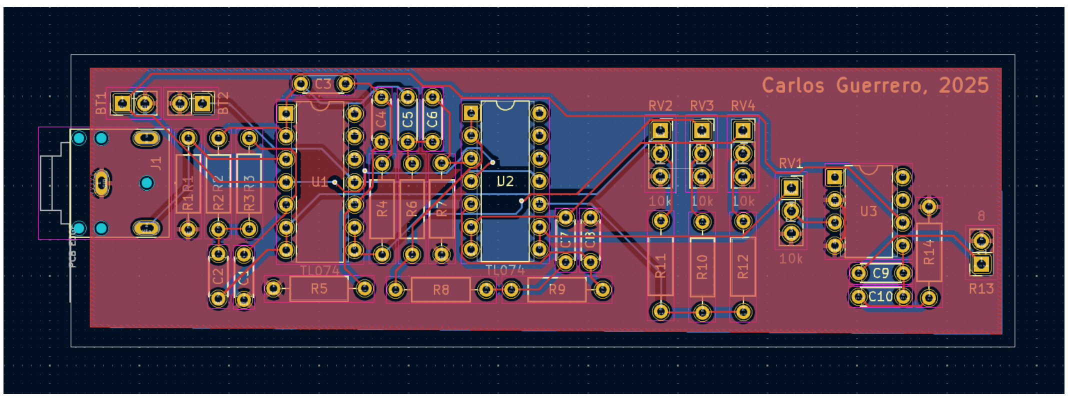

Figure 10: The final 2-layer PCB layout in KiCad, showing component placement and the top-layer ground plane pour.

4.2 Fabrication Files

To prepare the design for manufacturing, a standard set of fabrication files were generated from the completed KiCad PCB layout. These files, known as Gerbers and an Excellon drill file, contain the precise geometric data required by a PCB manufacturer like PCBWay to produce the board.

The Gerber files were exported using KiCad's "Plot" tool, with the "Use Protel filename extensions" option enabled for compatibility. A separate Gerber file was generated for each critical layer of the board, including the top and bottom copper (`F.Cu`, `B.Cu`), solder mask (`F.Mask`, `B.Mask`), silkscreen (`F.SilkS`), and the board outline (`Edge.Cuts`).

Separately, the Excellon drill file was generated to specify the location and size of all holes to be drilled in the board. All of these output files were then collected and compressed into a single `.zip` archive, creating a complete fabrication package ready for upload to the manufacturer.

4.3 Final Bill of Materials (BOM)

The complete list of components required for the fabrication of the audio equalizer circuit is detailed in Table 1.

Table 1: Bill of Materials

Quantity	Part / Value	Description
Integrated Circuits		
2	TL074	Quad JFET-Input Op-Amp
1	LM386	Low-Power Audio Amplifier
Resistors (1/4W)		
1	100 kΩ	Preamp Input Pull-down
4	10 kΩ	Summing Amplifier Resistors
2	6.2 kΩ	Bass / Mids Filter Resistors
2	3.9 kΩ	Treble / Mids Filter Resistors
Capacitors		
1	220 μF	LM386 Output Coupling (Electrolytic)
1	10 μF	LM386 Gain (Optional, Electrolytic)
2	0.1 μF (100nF)	Bass / Mids Filter (Ceramic/Film)
2	0.01 μF (10nF)	Treble / Mids Filter (Ceramic/Film)
Potentiometers		
4	10 kΩ Audio Taper	Bass, Mids, Treble, & Master Volume
Hardware / Miscellaneous		
1	3.5mm TRS Jack	Stereo Audio Input
1	8Ω Speaker	Small Project Speaker
2	9V Battery Clip	Power Connectors

5 Conclusion

5.1 Project Summary

This project successfully demonstrated the complete design, simulation, and implementation process for a three-band analog audio equalizer with an integrated power amplifier. All initial objectives outlined for the project were successfully met.

A stable, multi-stage analog circuit was designed utilizing TL074 operational amplifiers for the signal processing stages and an LM386 IC for the final power amplification. The performance of the circuit was thoroughly verified using LTspice. AC analysis confirmed that the Sallen-Key filters achieved their target cutoff frequencies, and transient analysis proved the circuit's ability to process complex audio signals without instability or significant distortion.

Finally, the validated schematic was fully translated into a two-layer printed circuit board (PCB) layout in KiCad, and a complete package of Gerber and drill files was generated. The resulting design is a complete and functional audio processing solution, ready for fabrication and physical assembly.

5.2 Challenges and Future Work

Challenges Encountered

A primary challenge during this project was diagnosing unexpected simulation behavior in LTspice. The initial oscillation of the active filters, for instance, was traced back to the op-amp models being unpowered due to missing power supply net labels—a non-obvious issue that highlights the importance of correctly modeling all circuit conditions. Another challenge involved understanding the practical difference between signal-level operational amplifiers and dedicated power amplifiers, particularly regarding the need for high current gain and handling DC offsets in single-supply designs.

Future Work and Potential Improvements

While this project successfully meets its objectives, several avenues exist for future expansion and improvement:

- **Stereo Version:** The most direct expansion would be to create a true stereo equalizer by duplicating the entire mono signal path for a second channel, creating a dual-ganged control system.
- **Expanded Frequency Control:** The design could be expanded into a 5- or 7-band graphic equalizer. This would involve designing additional band-pass filters with narrower bandwidths to provide more granular control over the audio spectrum.
- **True Bypass Switch:** A bypass switch could be implemented to route the input signal directly to the power amplifier, allowing for an instant A/B comparison between the original and the equalized signal.
- **Physical Prototyping:** The logical next step is to fabricate the designed PCB, assemble the components, and perform real-world testing with audio measurement equipment to compare its physical performance against the simulation results.

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